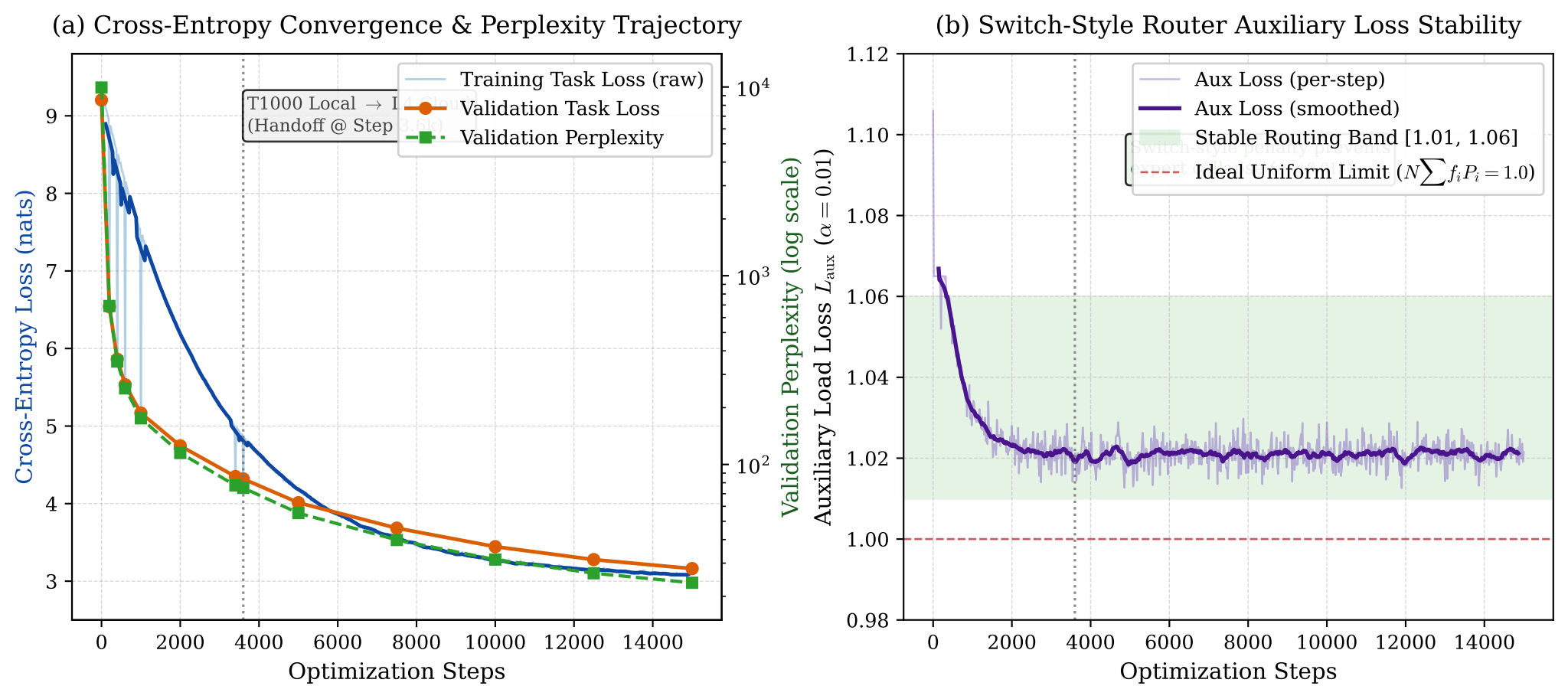

(a) Cross-Entropy Convergence & Perplexity Trajectory
Training Task Loss (raw)
Validation Task Loss
Validation Perplexity
Cross-Entropy Loss (nats)
Validation Perplexity (log scale)
Optimization Steps
(b) Switch-Style Router Auxiliary Loss Stability
Aux Loss (per-step)
Aux Loss (smoothed)
Stable Routing Band [1.01, 1.06]
Ideal Uniform Limit ($N\sum f_i P_i = 1.0$)
Auxiliary Load Loss L_{aux} ($\alpha = 0.01$)
Optimization Steps